Lab 3

Environmental Analysis in R

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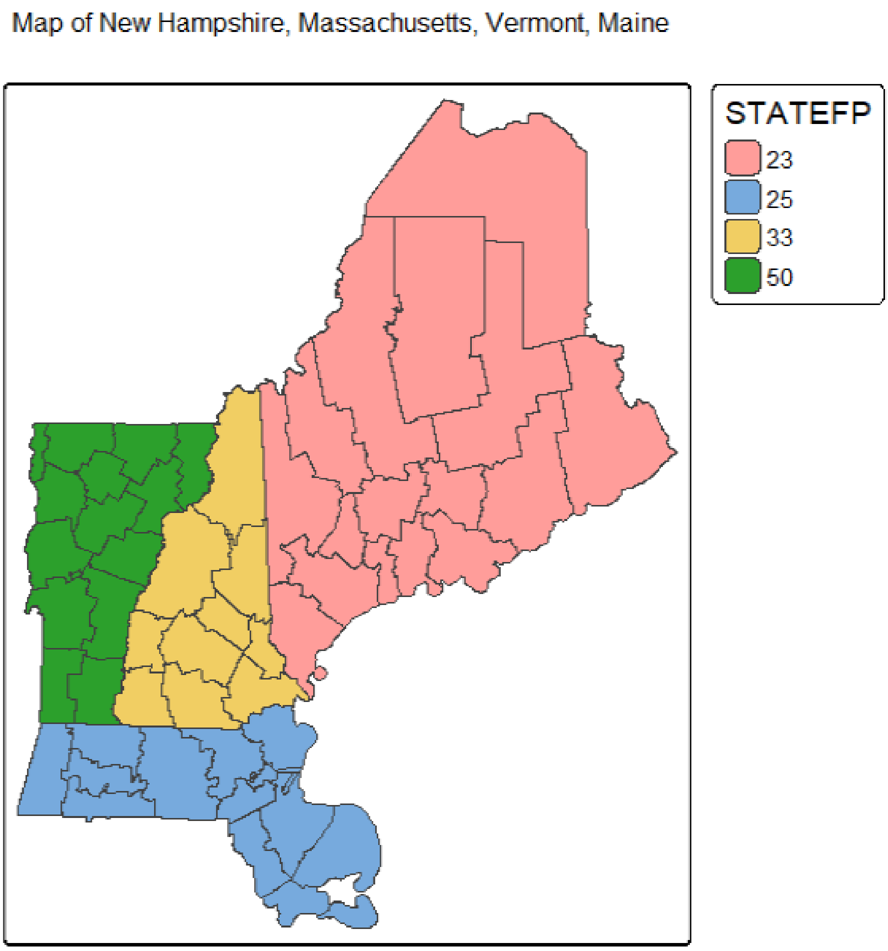
Task- 1

```
> # Task 1: get four contiguous states
> states4 <- d.joined %>% dplyr::filter(STATEFP == "50"|STATEFP == "33"|STATEFP == "23"|STATEFP == "25")
> write_sf(states4, "./data/states4.shp")
There were 50 or more warnings (use warnings() to see the first 50)
> tm_shape(states4) +
+   tm_borders() +
+   tm_polygons(fill = "STATEFP") +
+   tm_title("Map of New Hampshire, Massachusetts, Vermont, Maine") #plotting the spatial subset of the US
```



Task-2

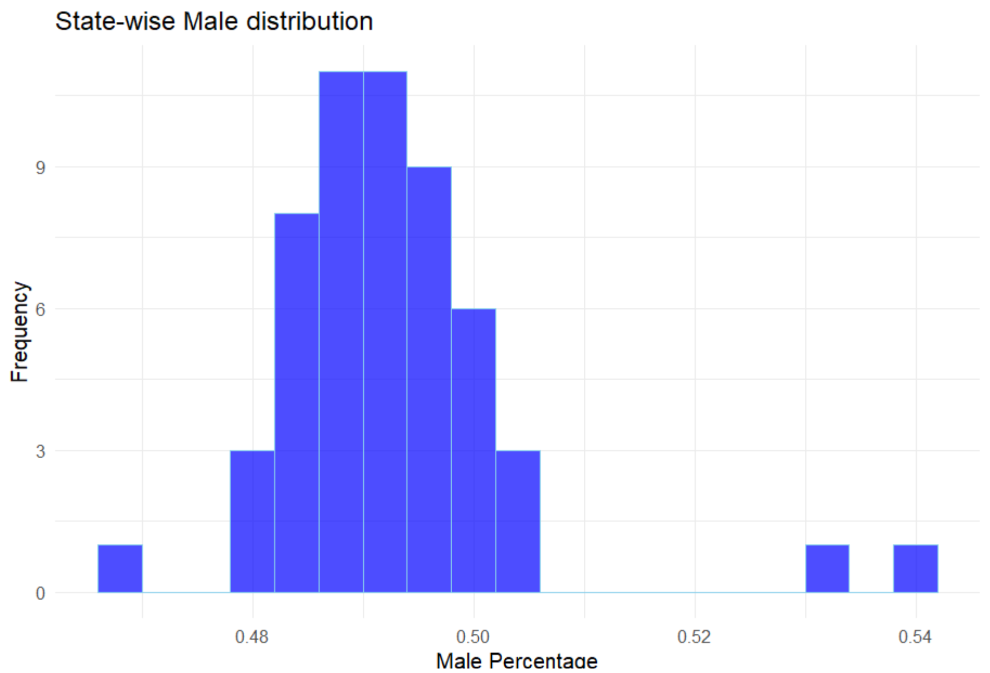
Here,

B01001e1= Total population, B01001e2= Total Male population, B01001e26= Total Female population [COUNTY_METADATA_2020]

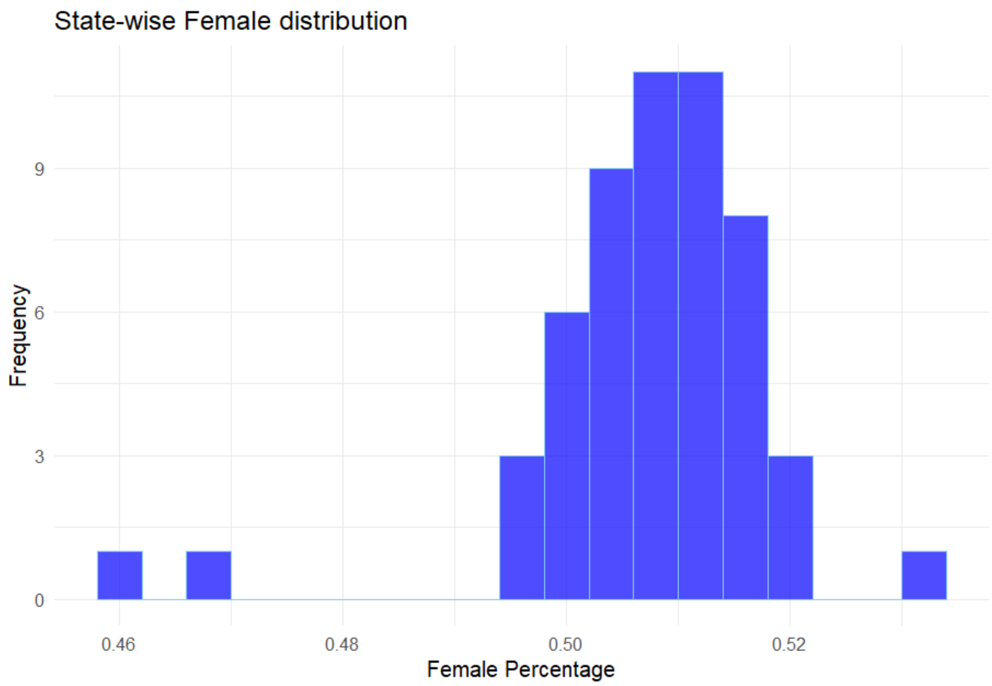
```
> #Task 2: Normalizing Male and Female population by Total population
> states4.population <- states4 %>%  mutate(.,MalePercent=B01001e2/B01001e1, FemalePercent=B01001e26/B01001e1)
```

Task-3

```
> #Task 3: Creating histogram of normalized male and female population
> states4.population %>% ggplot(., aes(x = MalePercent)) +
+     geom_histogram(binwidth= 0.004, fill= "blue", color = "skyblue", alpha = 0.7) +
+     theme_minimal() +
+     labs(title = "State-wise Male distribution",
+         x = "Male Percentage",
+         y= "Frequency")
```



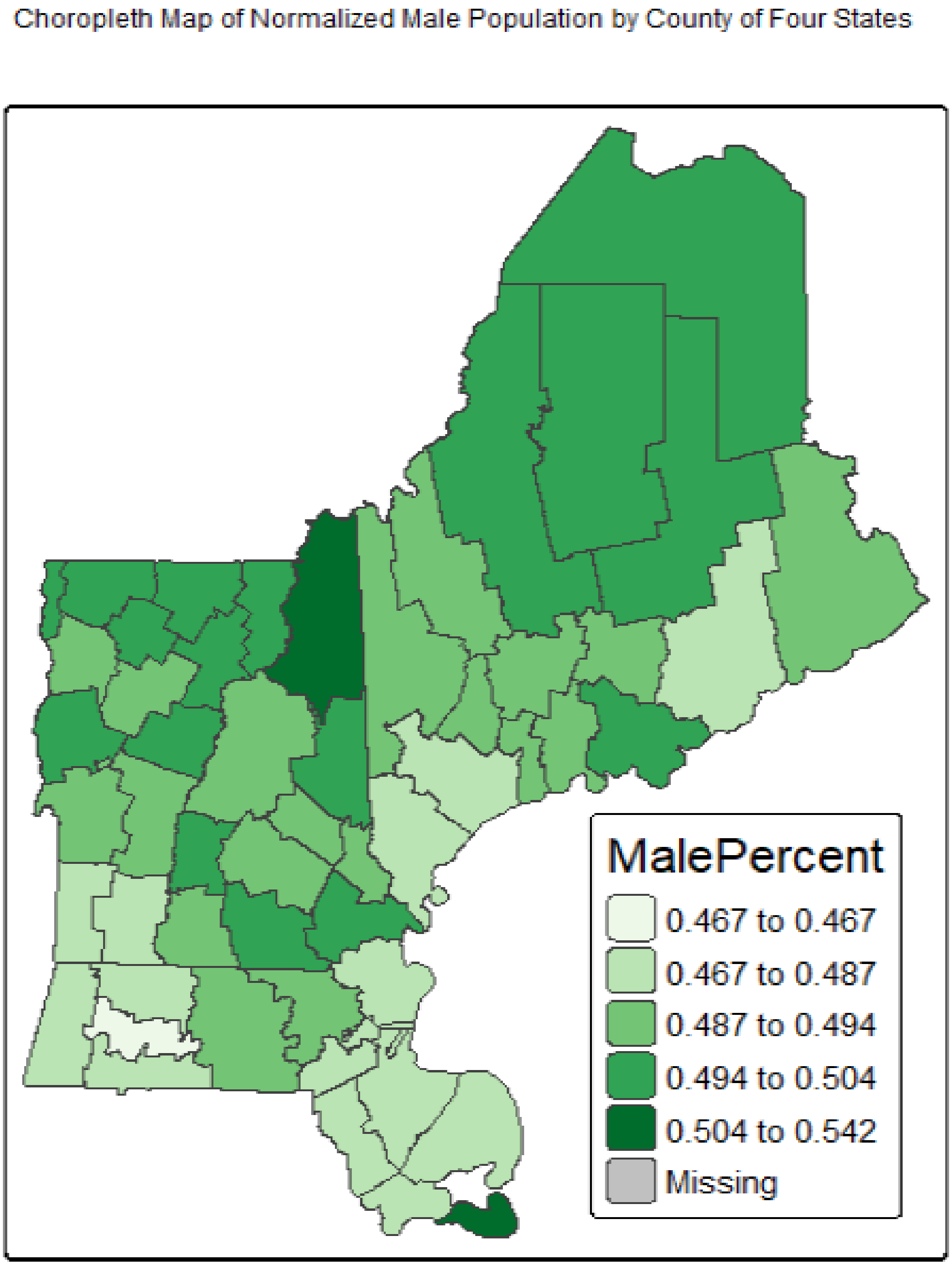

```
> #Or,
> states4.population %>% ggplot(., aes(x = FemalePercent)) +
+   geom_histogram(binwidth= 0.004, fill= "blue", color = "skyblue", alpha = 0.7) +
+   theme_minimal() +
+   labs(title = "State-wise Female distribution",
+        x = "Female Percentage",
+        y= "Frequency")
```



Task-4

```
> #Task 4: Creating choropleth map using the Jenks (Natural Breaks) classification scheme as it works
well for non-uniformly distributed data.
> #I found it most suitable after trying different schemes such as quantile, equal interval
> tm_shape(states4.population) +
+   tm_borders() +
+   tm_polygons(fill = "MalePercent", style = "jenks", palette = "brewer.greens") +
+   tm_title("Choropleth Map of Normalized Male Population by County of Four States") +
+   tm_layout(legend.title.size = 1, legend.position = c("right", "bottom"))

── tmap v3 code detected ───────────────────────────────────────────────────
[v3->v4] `tm_polygons()`: instead of `style = "jenks"`, use fill.scale =
`tm_scale_intervals()`.
ℹ Migrate the argument(s) 'style', 'palette' (rename to 'values') to
  'tm_scale_intervals(<HERE>)'
[plot mode] fit legend/component: Some legend items or map components do not fit
well, and are therefore rescaled.
ℹ Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```



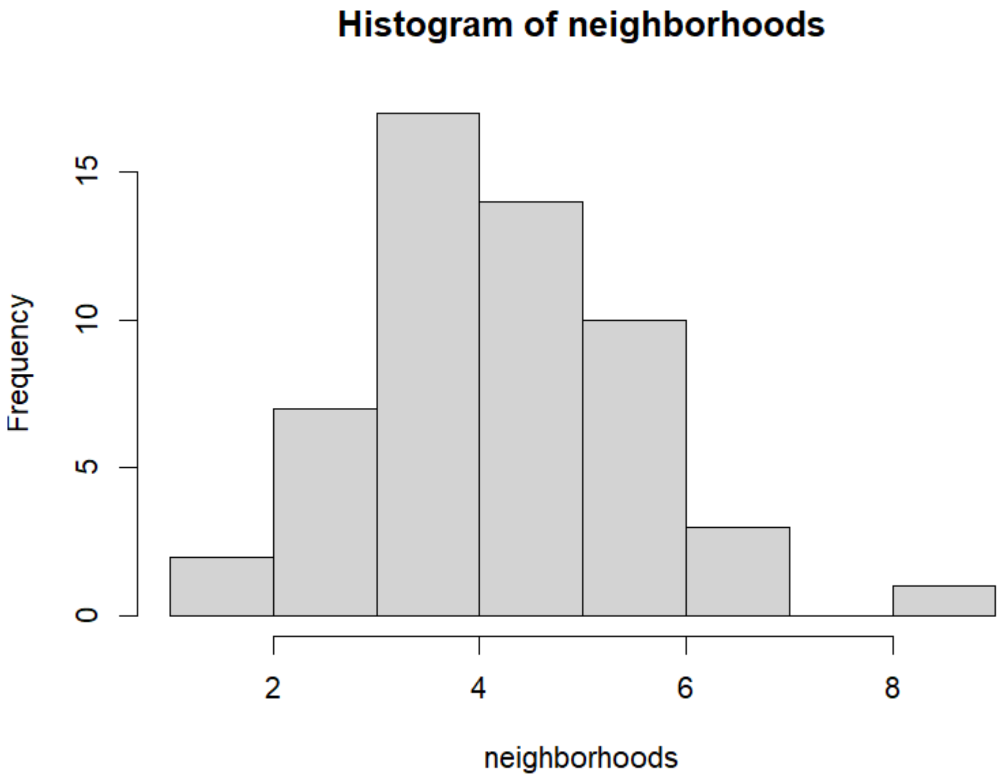
Task 5: I used the coordinate system mentioned in the Lab 3 material, which is ESRI:102010.

Task-5.1

```
> #5.1: Row-standardizing the W
> neighbor <- spdep::poly2nb(states4.projected, queen = TRUE)  #Calculate Queen's case neighbors
> lw <- nb2listw(neighbor, style="W", zero.policy=TRUE)
```

Task-5.2

```
> #5.2: Plotting a histogram of the number of neighbors
> neighborhoods <- attr(lw$weights,"comp")$d
> hist(neighborhoods)
```



Tak-5.3

```
> #5.3: Calculating the average number of neighbors
> avg_neighbors <- mean(neighborhoods)
> print(paste("Average number of neighbors (Queen contiguity):", avg_neighbors))
[1] "Average number of neighbors (Queen contiguity): 4.66666666666667"
```

Task-5.4

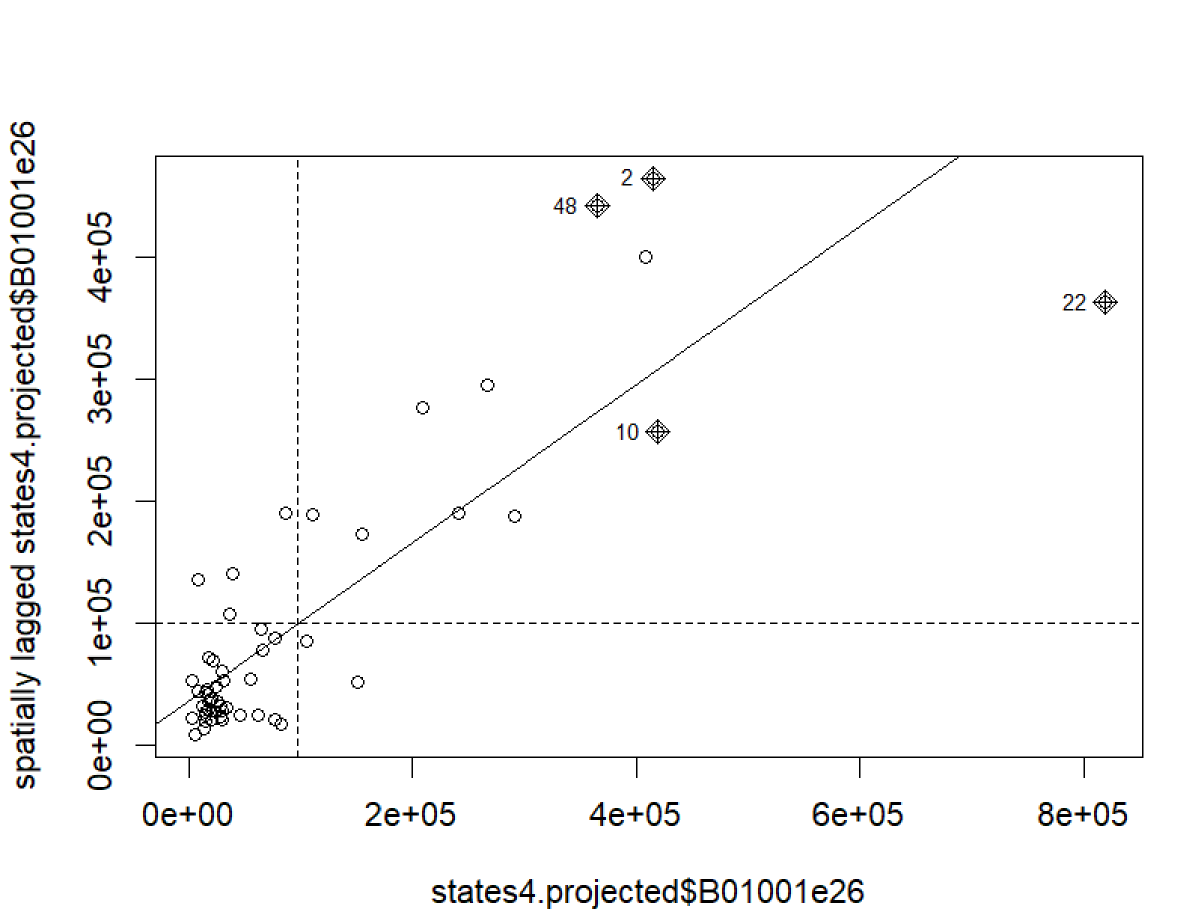
```
> #5.4: Making a Moran Plot using the var B01001e26[total female population]
> F.lag <- lag.listw(lw, states4.projected$B01001e26)
> moran.test(states4.projected$B01001e26, lw)

	Moran I test under randomisation

data:  states4.projected$B01001e26  
weights: lw    

Moran I statistic standard deviate = 8.2917, p-value < 2.2e-16
alternative hypothesis: greater
sample estimates:
Moran I statistic       Expectation          Variance 
      0.646518365      -0.018867925       0.006439653 

> moran.plot(states4.projected$B01001e26, lw, zero.policy=TRUE, plot=TRUE)
```



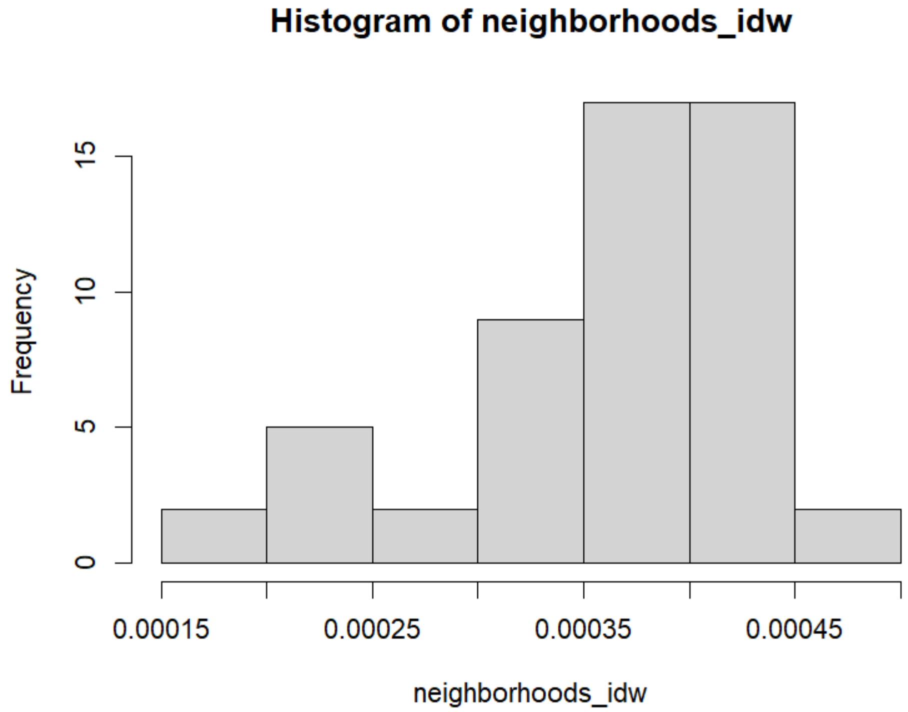
Task 6: The **"spdep"** document I followed to solve this task is added in the material folder.

Task- 6.1

```
> #6.1: Developing W using the IDW method
> #Checking and making the geometry compatible (calculating centroids) to create neighborhoods ba
sed on distance
> sf::st_geometry_type(states4.projected)
 [1] MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON
 [7] MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON
[13] MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON
[19] MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON
[25] MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON
[31] MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON
[37] MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON
[43] MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON
[49] MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON MULTIPOLYGON
18 Levels: GEOMETRY POINT LINESTRING POLYGON MULTIPOINT MULTILINESTRING ... TRIANGLE
> states4.centroid <- sf::st_centroid(states4.projected)
Warning message:
st_centroid assumes attributes are constant over geometries
> #Defining neighbors based on all distances (0 to infinity)
> neighbor_idw <- spdep::dnearneigh(states4.centroid, 0, Inf)
> lw_idw <- nb2listwdist(neighbor_idw, states4.centroid, type="idw", style="W", zero.policy=TRUE)
```

Task- 6.2

```
> #6.2: Plotting a histogram of the number of neighbors
> neighborhoods_idw <- attr(lw_idw$weights, "comp")$d
> hist(neighborhoods_idw)
```



Task- 6.3

```
> #6.3: Calculating the average number of neighbors
> avg_neighbors_idw <- mean(neighborhoods_idw)
> print(paste("Average number of neighbors (IDW):", avg_neighbors_idw))
[1] "Average number of neighbors (IDW): 0.000362309608861361"
```

Task- 6.4

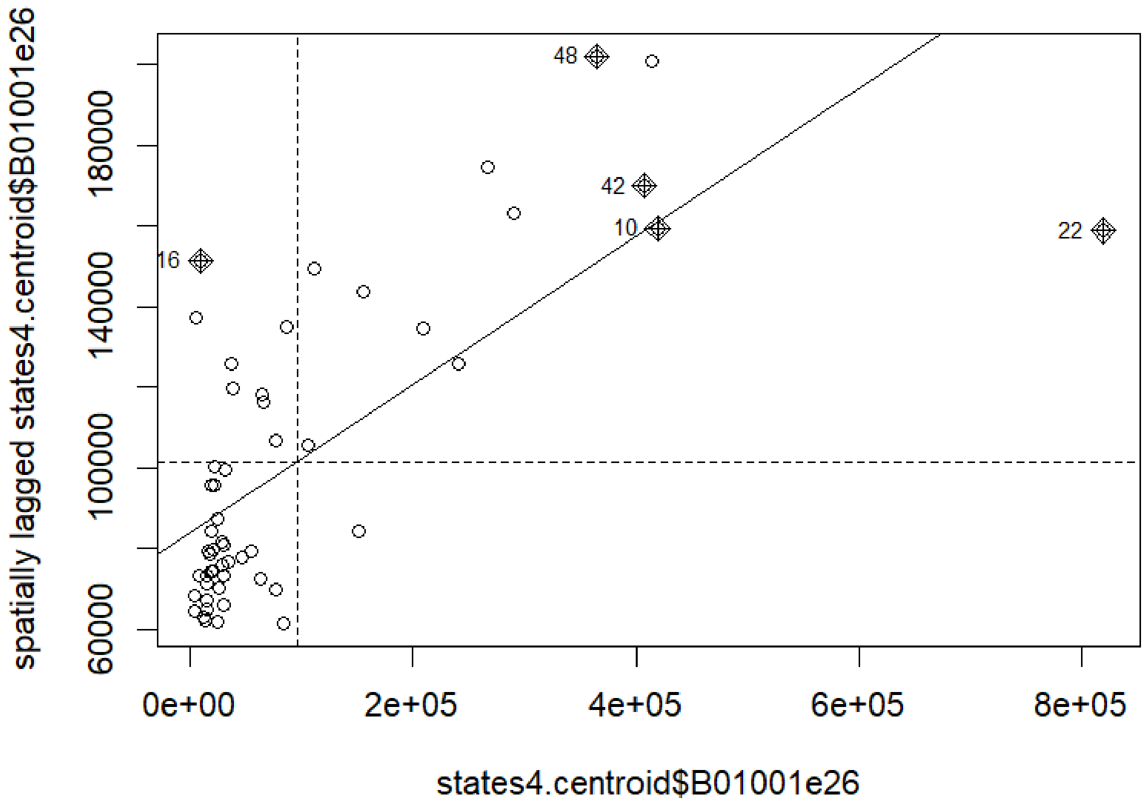
```
> #6.4: Making a Moran Plot using the var B01001e26 [total female population]
> F.lag.idw <- lag.listw(lw_idw, states4.centroid$B01001e26)
> moran.test(states4.centroid$B01001e26, lw_idw)

        Moran I test under randomisation

data:  states4.centroid$B01001e26
weights: lw_idw

Moran I statistic standard deviate = 11.448, p-value < 2.2e-16
alternative hypothesis: greater
sample estimates:
Moran I statistic       Expectation          Variance
     0.1838810032     -0.0188679245      0.0003136566

> moran.plot(states4.centroid$B01001e26, lw_idw, zero.policy = TRUE, plot = TRUE)
```



QA-1

Moran's I is a tool to analyze spatial autocorrelation. It indicates the degree to which a variable is correlated with itself across space. It also explains whether similar values are clustered together or dispersed geographically.

If I try to explain the calculation of Moran's I mentioned in the textbook, which is

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij}(z_i - \bar{z})(z_j - \bar{z})}{\sum_i (z_i - \bar{z})^2}$$

The weight between each pair of locations/polygons i and j is multiplied by each location's deviation from the mean values, and it is summed up. Then, it is divided by summing the squared deviations of the variable from the mean, which also reflects the overall variance of the variable. Finally, it is multiplied by the calculation of the total number of polygons or spatial units divided by the sum of the spatial weights.

In simple terms, to the best of my understanding, the formula generates an index value by comparing the values of each location and its neighbors and how they deviate from the average, providing weightage based on how close or connected those neighbors are and finally adding up all these comparisons. Here, the weights are defined as whether two regions are "neighbors" based on contiguity or considering the distance between them.

QA-2

A spatially lagged variable mainly represents the influence of neighboring locations on a particular variable of a location. It also refers to the weighted combination of the values of neighbors for that specific variable.

QA-3

Comparing the results I got from tasks 5 (defining weight based on contiguity) and 6 (defining weight based on distance), I found that although the expectation values for both tests are the same and Moran I statistics values are positive, the values of Moran's I, deviations, and variance are different. In both methods, there is a significant spatial autocorrelation, but the contiguity-based W indicates a strong positive relation and distance-based shows weaker relation.

For analysis, contiguity-based W should be used in those cases where relationships are expected to be influenced by directly adjacent areas sharing boundaries (queens or rooks). On the other hand, the IDW method should be used where influence is not restricted to direct or closest neighbors, and we want to understand gradual spatial patterns.

QA-4

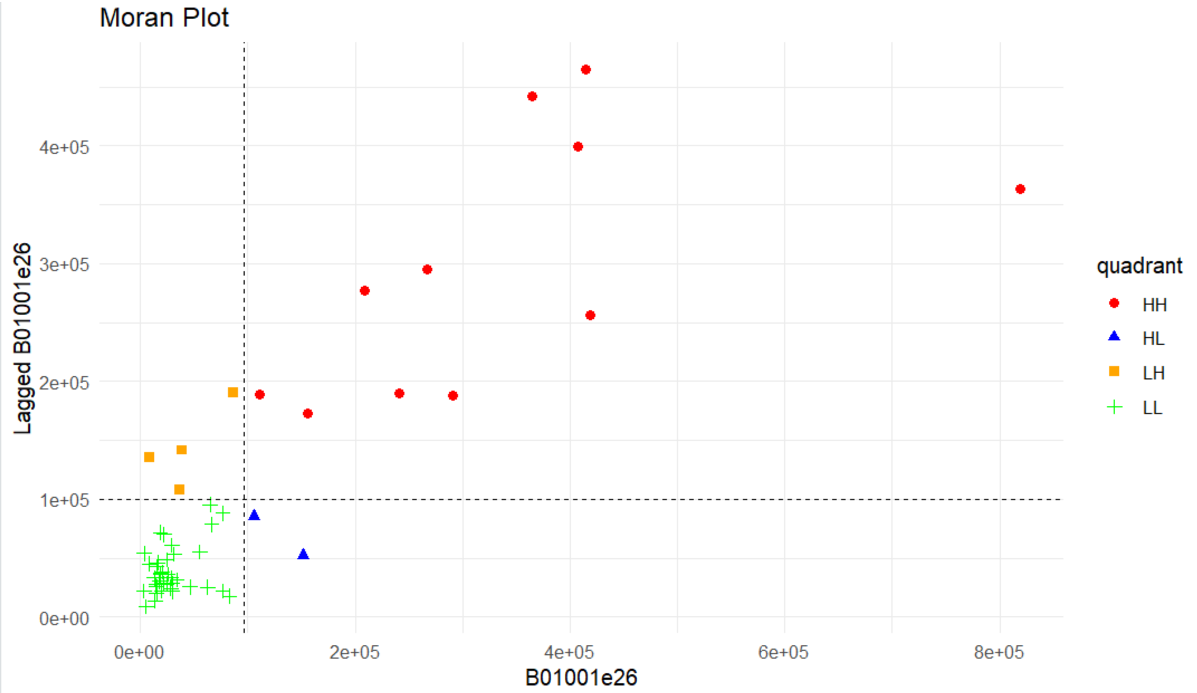
If an observation falls in the "H-L" quadrant, it represents the location with a high value compared to the values of its neighboring location whose values are low. In other words, high values above average are surrounded by below-average values. It helps us to detect whether there is any spatial outlier or an unusual relationship in the data.

As we know, the foundation of spatial autocorrelation is Tobler's First Law of Geography, "Everything is related to everything else, but near things are more related than distant things.". The H-L observations violate this rule by showing dissimilarity. These locations might be some areas of concern and need special attention.

Task- B1

```
> #B1
> #Creating dataset with the previously used variable and lag values
> moran_df <- data.frame(
+   unique_id = states4.projected$GEOID,
+   B01001e26 = states4.projected$B01001e26,
+   lag = F.lag
+ )
> #Adding var to the dataset mentioning quadrants based on values
> moran_df <- moran_df %>% mutate( quadrant =
+                                     ifelse(B01001e26 > mean(B01001e26, na.rm = TRUE) & lag > mean(lag, na.rm = TRUE), "HH",
+                                            ifelse(B01001e26 > mean(B01001e26, na.rm = TRUE) & lag <= mean(lag, na.rm = TRUE), "HL",
+                                                   ifelse(B01001e26 <= mean(B01001e26, na.rm = TRUE) & lag <= mean(lag, na.rm = TRUE), "LL", "LH"))))
> #Checking new dataset
> moran_df %>% head(5)
  unique_id B01001e26       lag quadrant
1     25015     86078 190389.75       LH
2     25025    414606 464682.25       HH
3     50011     24773  28146.25       LL
4     23001     55120  55386.60       LL
5     23023     18226  71697.75       LL
```

```
> #Creating Moran plot
> ggplot(moran_df, aes(x =B01001e26, y =lag , color = quadrant, shape = quadrant)) +
+   geom_point(size = 2) +
+   scale_color_manual(values = c("HH" = "red", "HL" = "blue", "LL" = "green", "LH" =
"orange")) +
+   geom_hline(yintercept=mean(moran_df$lag), lty=2) +
+   geom_vline(xintercept=mean(moran_df$B01001e26), lty=2) +
+   labs(title = "Moran Plot", x = "B01001e26", y = "Lagged B01001e26") +
+   theme_minimal()
```



Task- B2

```
> #B2
> joined.tables <- states4.projected %>% left_join(., moran_df, by = c("GEOID" = "unique_id"))
>
> tm_shape(joined.tables) +
+   tm_borders() +
+   tm_polygons(fill = "quadrant") +
+   tm_title("Choropleth Map of Moran Plot of the Counties of Four States")
[plot mode] fit legend/component: Some legend items or map components do not fit well, and are
therefore rescaled.
i Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```

Choropleth Map of Moran Plot of the Counties of Four States